**MODEL QUESTION PAPER**

TED(2010)5082 Reg. No…………………….

Revision 2010 Signature……………………

**SIXTH SEMESTER DIPLOMA EXAMINATION IN CHEMICAL ENGINEERING**

RUBBER TECHNOLOGY

(Maximum marks: 100) [*Time : 3 hours*

**PART-A** *Marks*

(Answer all questions in one or two sentences. Each carries 2 marks)

I

1 List any two applications of synthetic rubber.

2 What is a monomer.

3 State the role of activators in rubber processing.

4 List two extruded products.

5 Define cellular rubber. (5x2=10)

**PART-B**

II (Answer any five questions. Each carries 6 marks)

1 What are the major sources of natural rubber and state its importance. 6

2 Explain centrifuging of latex with flow chart. 6

3 Define elastomers with reference to extensibility and recovery. 6

4 Explain the properties of polyurethane. 6

5 List out the compounding ingredients. 6

6 What is a base polymer, explain its functions. 6

7 Explain compounding and production of battery box containers. 6

**PART-C**

(Answer one full question from each module. Each question carries 15 marks)

MODULE-I

| | | | |
|---|---|---|---|
| III | | | |
| | 1. | Describe the composition of NR latex and state the necessity of preservation of latex. | 8 |
| | 2. | Describe De-level centrifuge with a diagram. | 7 |
| | | OR | |
| IV | 1 | Explain the relevance of grading of sheet rubber. | 8 |
| | 2 | Differentiate crepe rubber and tyre rubber. | 7 |
| | | MODULE-II | |
| V | 1 | Define elastomers with reference to glass transition temperature. | 8 |
| | 2 | Explain the properties of silicon rubber. | 7 |
| | | OR | |
| VI | 1 | Explain the production of ethylene. | 8 |
| | 2 | What is polysulphide rubber? List its applications | 7 |
| | | MODULE-III | |
| VII | 1. | Distinguish CV, EV, SEV. | 8 |
| | 2. | Describe degradation and list the factors affecting degradation. | 7 |
| | | OR | |
| | 1. | Explain radiation vulcanization. | 7 |
| | 2. | Explain curing by fluidized bed. | 8 |
| | | MODULE-IV | |
| IX | 1. | Define cellular rubber and classify different cellular rubber. | 8 |
| | 2. | What are the defects in the manufacturing of rubber tubing. | 7 |
| | | OR | |
| X | 1. | What is a calendared product? List out various calendared products. | 8 |
| | 2. | State the principle of creating cellular structure in rubber. | 7 |